MENTAL HEALTH PREDICTION USING MACHINE LEARNING APPROACH

By

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STUDENT DECLARATION

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Signed,

Vijeth Balasubramaniam

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ABSTRACT

The mental healthcare industry has always worked in conjunction with mental health datasets to support clinical judgment. Traditionally, human expertise has been the primary tool in diagnosing and treating mental health conditions. However, with technological advancements in artificial intelligence (AI) and machine learning (ML), this gap is narrowing. Machine intelligence now empowers professionals with cognitive decision-making capabilities, complementing human judgment. In particular, the integration of ML algorithms into medical datasets has enabled more efficient and accurate clinical decision-making processes.

Among the many health concerns facing society today, mental health diseases stand out as a major contributor to mortality and disability worldwide. They are a leading cause of suffering and premature death in many countries. As the demand for mental healthcare continues to rise, there is an urgent need for more innovative solutions. This prediction model focuses on the early identification of mental health risks, specifically among students. The aim is to address the increasing prevalence of mental health issues in younger populations, providing valuable insights for early intervention.

This model leverages multiple machine learning classification algorithms, such as Logistic Regression, Decision Trees, Random Forests, Gradient Boosting, AdaBoost, and K-Nearest Neighbors. These algorithms are applied to student data, including features such as academic performance, alcohol consumption, health status, and absenteeism, which are used as predictors of mental health risk. By analyzing patterns in this data, the model can predict whether a student is at risk of developing mental health issues, helping stakeholders make informed decisions.

The performance of the model is evaluated through 30 runs, using ROC-AUC scores and accuracy metrics to measure its predictive power. Additionally, a hypothetical scenario tests the model's robustness by removing certain features to examine how they affect the predictions. The study demonstrates the potential of machine learning to assist in early detection and intervention in mental health, highlighting its role in shaping the future of healthcare decision-making.

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Chapter 1. INTRODUCTION

1.1 Scope

This dissertation's focus is on using machine learning approaches to identify simulated students' mental health risks based on behavioural and academic data[3]. It evaluates the effectiveness of machine learning algorithms like Gradient Boosting, AdaBoost, Decision Trees, Random Forests, and K-Nearest Neighbours in identifying potential mental health issues such as stress, anxiety, and depression. These algorithms have previously been used in other studies, but to date, no study has compared them all in a single study across a common dataset. However, the study does not address clinical diagnosis or treatment[5]. Instead, it emphasizes risk prediction based on available data points. The scope is limited to data-driven predictions.

1.2 Problem Statement

Mental health issues among students are increasing, driven by academic pressure, social challenges, and lifestyle factors[6]. Traditional mental health assessments are subjective, time-consuming, and reliant on human judgment, making early detection difficult. This can lead to severe outcomes like self-harm, chronic mental illness, or student dropouts[9]. This dissertation addresses how machine learning could be used to accurately predict mental health risks in students by comparing the various model's performance on the same dataset, such a system could, in principle, be used to help educational institutions implement early interventions and provide targeted mental health support. However, there are ethical concerns in using real data, therefore this study uses a proxy classification approach for labelling training and test instances.

1.3 Approach

This dissertation adopts a structured approach to building a machine learning-based prediction model for assessing mental health risks[1]. The process begins with data collection and preprocessing, focusing on factors like lifestyle, attendance, and academic performance. Multiple classification algorithms are trained and optimized using techniques such as Grid Search to achieve high predictive performance[3]. The models are evaluated using metrics like accuracy and ROC-AUC scores, with further feature selection and model optimization to improve robustness. A comparative analysis identifies the most effective algorithm for mental health risk prediction[7].

1.4 Outcome

A machine learning model that can forecast the simulated mental health outcome based on behavioural and academic data is the dissertation's anticipated output. The results will determine the best algorithm for predicting possible mental health risks, which could enhance how educational institutions keep an eye on students' wellbeing. The approach demonstrates that the use of machine learning (if used with real data) could be used to lessen the detrimental effects of mental health illnesses on student performance and general well-being based on the classification of a set of features of students by providing early warnings, which could assist administrators and mental health experts in providing timely interventions.

Chapter 2. BACKGROUND AND REVIEW OF LITERATURE

2.1 Related Work

In recent years, there has been a significant increase in interest in the application of machine learning (ML) techniques for mental health prediction. Several research studies have investigated how well these algorithms work in detecting mental health problems in a variety of populations, especially students, where prompt interventions depend on early-diagnosis.

In a study presented by Rahman et al. (2023), machine learning models are used to predict mental well-being based on health behaviour data from college students[1]. Their study highlights the significance of behavioural and lifestyle factors in forecasting mental health outcomes. The study uses a number of machine learning methods, including Gradient Boosting and Random Forests, to predict mental health problems and demonstrates how well some health behaviours can function as predictors. In recent years, there has been a significant increase in interest in the application of machine learning (ML) techniques for mental health prediction. This research closely resembles the current study, which uses comparable machine learning techniques to predict mental health risks among students by looking at behavioural and academic data.

Sahu and Debbarma (2023) employ a range of machine learning approaches, such as Support Vector Machines (SVM), Decision Trees, and K-nearest neighbors (KNN), to precisely predict students' mental health[2]. Their study emphasizes how crucial feature selection and preprocessing are to maximise the effectiveness of machine learning models. By showing that combining machine learning with education-based data can greatly enhance mental health risk forecasts and strengthen the efficacy of classification algorithms, this work adds to the expanding body of research.

By utilising machine learning models to forecast mental health in a higher education setting, Mutalib et al. (2021) expand on this work. They assess the effectiveness of several algorithms and stress the significance of model optimisation as well as the difficulties associated with imbalanced datasets[3]. Their research emphasises how machine learning models must be continuously improved in order to appropriately detect mental health concerns, which is a major focus of the current dissertation.

Chung and Teo (2022) offer an extensive analysis of models for predicting mental health, classifying machine learning techniques and their uses in a range of fields. Their research highlights the difficulties of using machine learning to mental health, such as problems with algorithmic bias, data privacy, and the complexity of human emotions[4]. This dissertation acknowledges these difficulties, yet it nonetheless focusses on applying machine learning models to early risk prediction in a learning environment.

Furthermore, Vaishnavi et al. (2022) investigate the use of machine learning algorithms to forecast mental health conditions. Their study examines many variables that affect mental health outcomes and emphasises the effectiveness of various models, especially when managing enormous datasets[6]. By highlighting the significance of algorithmic selection and data pretreatment in enhancing model accuracy, this work enhances the current study.

The Alan Turing Institute has also investigated the application of artificial intelligence (AI) to precision mental health, emphasising the use of AI to provide more precise and tailored mental health services. The significance of predictive modelling in mental health is highlighted by their research, especially in terms of spotting early warning indicators for more successful interventions[5].

Lastly, Baba and Bunji (2022) concentrate on the application of machine learning and yearly

student health surveys to forecast mental health issues. Their results support the goal of the current project, which is to use academic and behavioural data to forecast mental health risks[7]. Annual surveys, when paired with machine learning techniques, can yield reliable predictions of students' mental health difficulties.

2.2 Literature

Machine learning is being used more and more to predict mental health, especially in student populations, by identifying early hazards based on academic and behavioural data. Algorithms such as Random Forests, Decision Trees, and Gradient Boosting have proven to be useful in forecasting mental health outcomes from non-clinical data, as shown by recent studies. Important variables like lifestyle choices, health-related behaviours, and academic achievement have been shown to be reliable indicators. The difficulties with algorithmic bias, imbalanced datasets, and data privacy are also highlighted by a number of studies in this area. These studies highlight how machine learning can lower mental health risks, enable prompt interventions, and enhance wellbeing in learning environments. By utilising these insights, our study develops an enhanced prediction model that can help with early detection and intervention.

CHAPTER 3. DESIGN

3.1 Original Design

The primary goal of this project is to predict mental health issues among students using machine learning algorithms. In the design phase, the proposed testing of various models, including Gradient Boosting, Decision Trees, and SVM. The dataset for the project contained 649 instances, with features such as student absences, failures, health status, and alcohol consumption.

In the original design, the following key components were outlined:

- **Data Preprocessing**: Cleaning and transforming the data, handling categorical variables via OneHotEncoding.
- **Model Selection**: Implementing multiple machine learning models—such as Decision Trees, Gradient Boosting, and SVM—and comparing their performance using metrics like accuracy, precision, recall, and F1-score.
- **Model Evaluation**: Utilizing metrics like ROC-AUC, precision, recall, and specificity to identify the most efficient model.
- **Tools**: Python programming language and libraries, including Scikit-learn, Seaborn, and Matplotlib.

3.2 Changes to Original Design

During the implementation phase, several updates were applied based on testing and evaluation results. An explanation of how each changes influences the model's performance is explained below,

1. **Data Preprocessing Enhancements**:

   The original design lacked a clear target definition for mental health issues, which could lead to data leakage. The updated preprocessing ensured data integrity and enhanced predictive accuracy by creating a specific target column (mental_health_issues) based on relevant features and dropping those features afterward.

2. **Model Training and Hyperparameter Tuning**:

   The original plan limited the models and didn't focus on optimization. The updated design included additional models like **Logistic Regression** and **Random Forest** and applied **Grid Search** for hyperparameter tuning. This optimized the models for better performance and ensured robustness with 30 independent runs.

3. **Extended Metrics**:

   Initially, only basic metrics like accuracy and precision were used, which didn't fully assess model performance. Adding **specificity** and plotting **ROC-AUC scores** addressed this by providing a more detailed evaluation, particularly for negative case identification, leading to a more balanced performance assessment.

4. **Visualization and Model Comparisons**:

   The original plan relied on basic ROC-AUC curves, which lacked depth. By incorporating **bar plots with error bars** and **Gaussian distribution plots**, the updated approach allowed for a clearer visualization of model stability and performance consistency across multiple runs.

5. **Feature Reduction (Hypothetical Scenario)**:

   The original design didn't consider feature sensitivity. The updated version introduced a feature reduction test, removing Medu, Fedu, and studytime to evaluate model robustness. This allowed for a better understanding of which models could generalize well with reduced data.

6. **Comparison Between Scenarios**:

   The original plan didn't include a scenario comparison. The updated implementation introduced a comparison between the original and feature-reduced datasets, revealing how the removal of features affected model performance, particularly ROC-AUC scores, highlighting models' adaptability.

CHAPTER 4. DATASET

4.1 Ethical use of data

The dataset used in this project is freely accessible through the UCI Machine Learning Repository, which offers a range of datasets for scholarly and scientific uses. The dataset utilised in this investigation comprises 649 occurrences pertaining to student performance, encompassing characteristics such as parental education, study time, alcohol usage, and additional aspects that may have an effect on mental health.

The dataset used in this project consists of several attributes that capture various aspects of student performance and background. These include school (categorical, representing the school attended), sex (categorical, indicating gender), and age (numeric, ranging from 15 to 22 years). Students' address (urban or rural) and family size are also recorded, along with parents' education levels, represented by Medu and Fedu, which range from 0 (no education) to 4 (higher education). The dataset captures parental jobs, reasons for school choice, and guardian information. Variables like study time, travel time, and failures represent academic-related data, while alcohol consumption on both weekdays (Dalc) and weekends (Walc) is provided on a scale of 1 to 5. Other behavioral data include participation in extra-curricular activities, family relations, free time, and romantic relationships. The dataset also records school absences, along with health and social behavior indicators such as internet access and going out with friends. This comprehensive set of attributes offers insights into the factors affecting students' mental health and overall academic performance.

The dataset is available for use, distribution, and alteration under the terms of the **Creative Commons Attribution 4.0 International (CC BY 4.0) license**, provided that the original authors are properly credited. As long as the source is appropriately cited in any papers or research outcomes resulting from the dataset's use, it is free to use for research purposes. In the paper "Using data mining to predict secondary school student performance" by Cortez and Silva, the dataset was used to predict student academic success by analyzing factors such as demographics, academic performance, and behavioral attributes. The authors applied data mining techniques, including decision trees, random forests, and support vector machines, to identify the most significant predictors of performance. The outcome demonstrated that factors like study time, family support, and parental education levels had the most significant impact on students' grades, providing insights into early identification of at-risk students.

Adhering to ethical principles is crucial when handling sensitive data, such as mental health indicators, to guarantee that the research does not cause harm to individuals or groups. Because the data in this instance has been anonymised and does not contain personally identifying information (PII), there is less chance that someone's privacy will be violated. Even so, it's critical to recognise the consequences of utilising this data for forecasting and to take care while analysing the results to prevent stigmatising any specific group.

Ethical questions of prejudice and fairness also surface in the context of machine learning and mental health prediction. In order to prevent bias, machine learning models need to be created and verified, especially when it comes to sensitive characteristics like gender or socioeconomic status. To guarantee equity and minimise any inadvertent bias or detrimental inferences made from the information, audits and assessments of the models must to be carried out on a regular basis.

Efforts have been made as part of this project to ensure adherence to ethical standards, such as openness in data preprocessing and guaranteeing that the model's results are applied to assist early intervention strategies for mental health issues, potentially benefiting students, instead of being used for discriminatory or punitive measures.

4.2 Data Preprocessing

Data preprocessing is a crucial step in machine learning projects, as it ensures that the data is in a clean and appropriate format for the models to process[6]. In this project, the preprocessing phase involved handling both numerical and categorical variables, creating a target variable for prediction, and transforming the data into a standardized format for model training[15].

Key Steps in Data Preprocessing:

1. **Creation of the Target Variable:**

   The target variable, **mental_health_issues**, is created by logically combining several important features from the dataset, including **failures**, **absences**, **health status**, and **alcohol consumption**. This process helps classify whether a student is likely to face mental health challenges based on these attributes. After the target variable is created, the original features used in its formation are removed to avoid redundancy and potential data leakage, as including them during model training would skew results by allowing the model to predict the target using the same variables it was derived from. The challenge in this step was deciding which features best represent mental health risks, as the dataset does not provide explicit labels for mental health, making this a crucial assumption-based transformation.

```
# Create Target Column based on assumed criteria
df['mental_health_issues'] = ((df['failures'] > 1) |
                              (df['absences'] > 10) |
                              (df['health'] < 3) |
                              (df['Dalc'] > 2) |
                              (df['Walc'] > 2)).astype(int)
```

Snippet 1

2. **Handling Categorical Variables:**

   Categorical variables, such as **school** and **sex**, are converted into numerical values using **one-hot encoding**. This transformation is essential because most machine learning algorithms require numerical input. One-hot encoding converts each category into separate binary columns, where 1 indicates the presence of a category and 0 represents its absence. The challenge here was the increase in the number of dimensions, as each categorical variable's categories translate into additional columns, leading to a potential curse of dimensionality if not handled carefully.

```
# Encode categorical variables using OneHotEncoding
categorical_columns = df.select_dtypes(include=['object']).columns
df_encoded = pd.get_dummies(df, columns=categorical_columns, drop_first=True)
```

Snippet 2

3. **Splitting Features and Target:**

   Once the preprocessing steps are completed, the dataset is split into **features (X)** and the **target variable (y)**. This step ensures that the independent variables (features) are used for training the model while the dependent variable (target) is used to evaluate the model's predictions. This straightforward task prepares the dataset for the next steps in the machine learning pipeline. One key challenge is ensuring that the dataset remains balanced to prevent skewed model performance when the target classes are imbalanced.

4. **Standardization:**

   Standardizing the features ensures that all numerical variables have a mean of 0 and a standard deviation of 1. This step is crucial for certain machine learning algorithms, such as **Logistic Regression** and **SVM**, which are sensitive to the scale of the input data. Without standardization, features with larger numerical ranges may disproportionately influence the model's predictions. The challenge here was ensuring that categorical variables were excluded from this process, as standardization only applies to numerical data.

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Snippet 3

CHAPTER 5. IMPLEMENTATION

5.1 Methodology

The research followed a structured methodology:

1. **Data Collection & Preprocessing**: Cleaned and transformed data to prepare it for machine learning algorithms.
2. **Model Implementation**: Various models were trained and optimized using **Grid Search** to achieve high accuracy.
3. **Evaluation**: Performance was evaluated over 30 runs with different data splits, and ROC-AUC was the main metric for comparing models. The robustness of the models was tested through feature reduction, and comparisons between original and feature-reduced datasets were made

5.2 Logistic Regression

Logistic Regression is a linear model that assumes a Gaussian distribution of the features, making it a natural fit when the data closely follows such a distribution. In this project, Logistic Regression consistently achieved the highest ROC-AUC (0.673) across multiple runs, indicating that it was particularly adept at distinguishing between students with and without mental health issues[2]. Its high recall (0.844) shows the model's effectiveness at capturing instances of mental illness, while the F1 score of 0.741 further highlights its balanced

performance[3]. The Gaussian distribution of the data supports this model's linear assumptions, helping it form clear decision boundaries that align well with the spread of data (fig. 1). Logistic Regression proves to be an excellent model for predicting mental health outcomes, particularly when dealing with well-distributed data, as seen in this study.

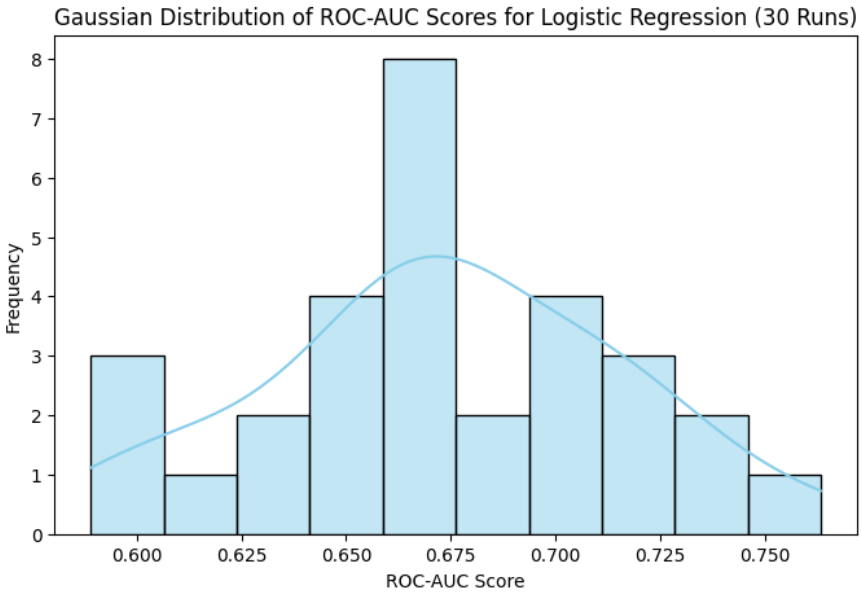


Figure 1

This figure shows the distribution of ROC-AUC scores for the Logistic Regression model. The scores are tightly clustered around the mean, indicating that Logistic Regression consistently performed well across multiple runs, with minimal variance in its predictions.

5.3 Decision Tree

Decision Tree does not rely on any underlying data distribution, such as Gaussian, and instead splits the data hierarchically based on feature values. In this project, it had a lower ROC-AUC (0.636) compared to Logistic Regression, with a slightly higher precision (0.671) but lower recall (0.774). This suggests that while the Decision Tree was effective at making precise predictions[5], it was less capable of identifying all instances of mental illness. The normal distribution of the data may not have favored the Decision Tree, as its non-parametric nature often leads to overfitting on certain splits of the data (fig. 2). Although it can capture complex relationships, the slightly lower performance indicates that it struggles with the Gaussian-like structure of the data, resulting in more variability in predictions.

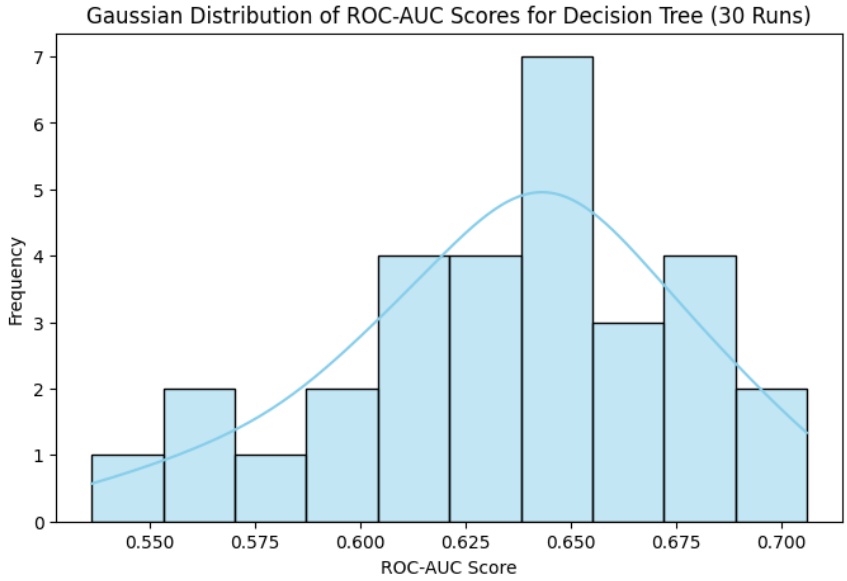


Figure 2

The ROC-AUC scores for the Decision Tree model exhibit a wider spread compared to Logistic Regression. This suggests more variability in the model's performance, possibly due to overfitting in certain runs.

5.4 Random Forest

Random Forest, as an ensemble of Decision Trees, averaged predictions across multiple trees, allowing it to better generalize to unseen data. It performed well in this project, with a balanced ROC-AUC of 0.662 and a respectable recall (0.802). The model's ability to handle various distributions without assumptions allowed it to generalize effectively, though its performance was not as high as Logistic Regression or AdaBoost[9]. Random Forest's reliance on averaging decisions helped mitigate the overfitting issues seen in Decision Trees, and the model handled the normal distribution of data competently (fig. 3). The moderate ROC-AUC score reflects its strength in balancing variance and bias, though it slightly underperformed on Gaussian-like data when compared to simpler models like Logistic Regression.

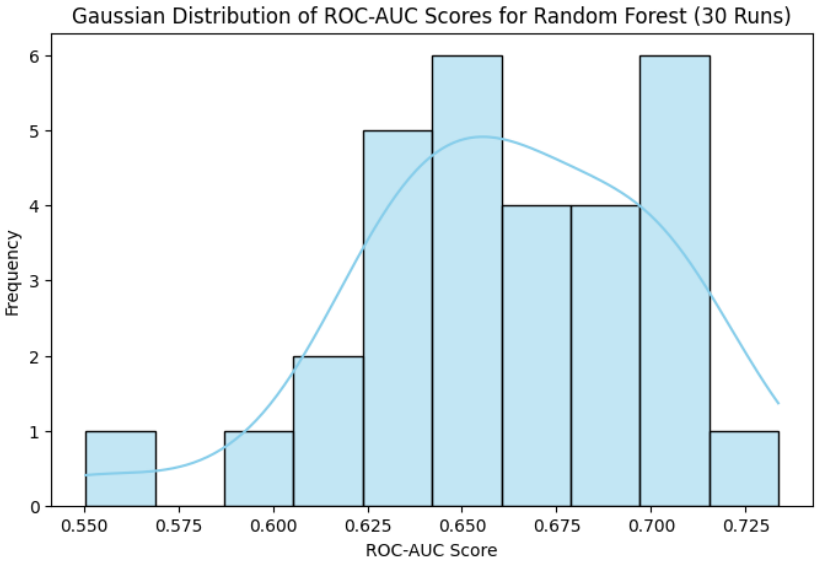


Figure 3

The Random Forest model's scores are more stable than those of the Decision Tree, as the distribution is narrower. This reflects the ensemble method's ability to generalize better, reducing overfitting and yielding more consistent results.

5.5 Support Vector Machine

Support Vector Machine (SVM)**,** with its capacity for separating classes through the use of kernel functions, performed reasonably well with a ROC-AUC of 0.655 and a precision of 0.677. SVM's reliance on finding an optimal hyperplane was challenged by the slightly non-linear separation required in this dataset, as the Gaussian distribution suggests that data points might not be perfectly separable[1]. SVM performed adequately with the RBF kernel, but its performance lagged behind Logistic Regression and AdaBoost due to its sensitivity to the data's structure (fig. 4). The results indicate that while SVM can model more complex relationships, its assumptions about the separation of classes are not fully suited to the Gaussian nature of this dataset.

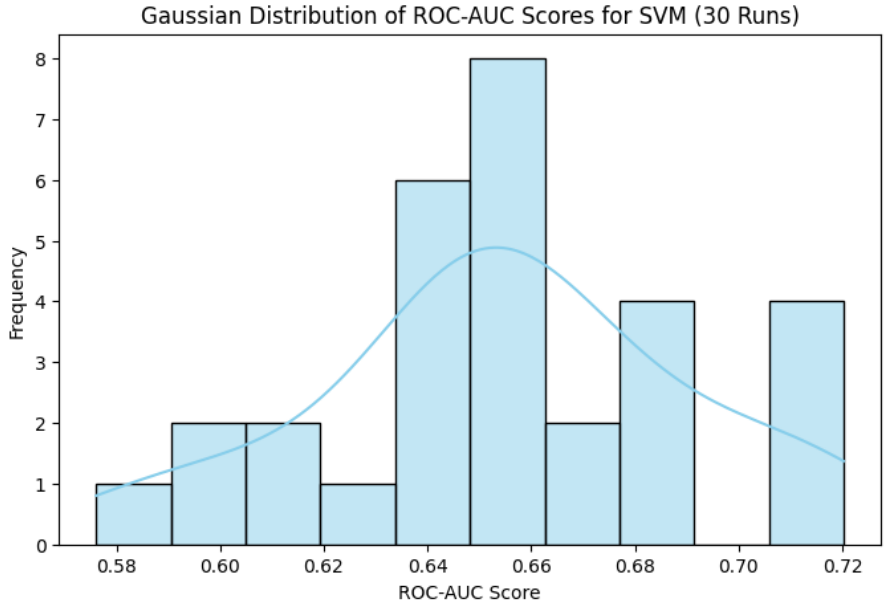


Figure 4

SVM's ROC-AUC scores are moderately spread, indicating that while the model performed well in some cases, it struggled in others. The distribution shows that SVM's performance was somewhat inconsistent across different runs.

5.6 Gradient Boosting

Gradient Boosting is another ensemble method that builds weak learners iteratively, focusing on correcting misclassified points at each stage[14]. In this project, it showed balanced performance with a ROC-AUC of 0.653, a precision of 0.653, and a strong recall of 0.828. Although Gradient Boosting didn't outperform Logistic Regression or AdaBoost, it handled the dataset's Gaussian distribution well, progressively improving the classification of instances[8]. The model's strength lies in its ability to fine-tune predictions across different stages, though the variability in the data's distribution meant that it performed slightly below its potential (fig. 5). Despite this, Gradient Boosting remained a solid model, demonstrating good recall and handling the dataset with consistency.

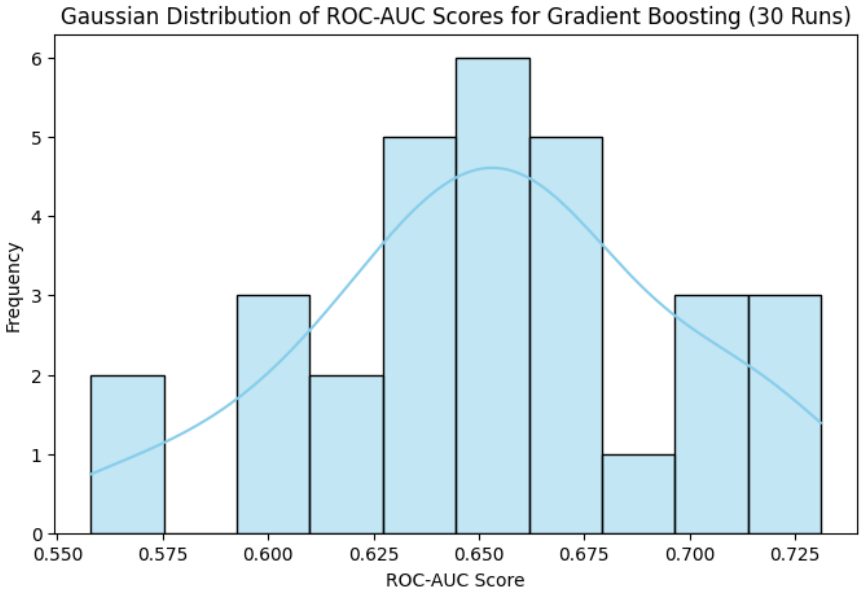


Figure 5

The Gradient Boosting model shows a relatively tight distribution of ROC-AUC scores, meaning its performance was stable across the runs. The model consistently improved classification by focusing on hard-to-predict instances.

5.7 AdaBoost

AdaBoost, like Gradient Boosting, iteratively corrects misclassified points but assigns higher weights to more difficult cases[12]. This allowed AdaBoost to achieve high results, with a ROC-AUC of 0.673 and the highest recall (0.846) alongside Logistic Regression. AdaBoost's ability to adapt to the data made it particularly well-suited for this dataset, where it leveraged the Gaussian distribution to progressively improve its predictions[15]. The tight performance consistency across multiple runs suggests that AdaBoost is robust and adaptable, making it one of the top performers for this dataset (fig. 6). The model's error correction mechanism proved highly effective in capturing the key factors influencing mental health outcomes, offering strong precision and recall.

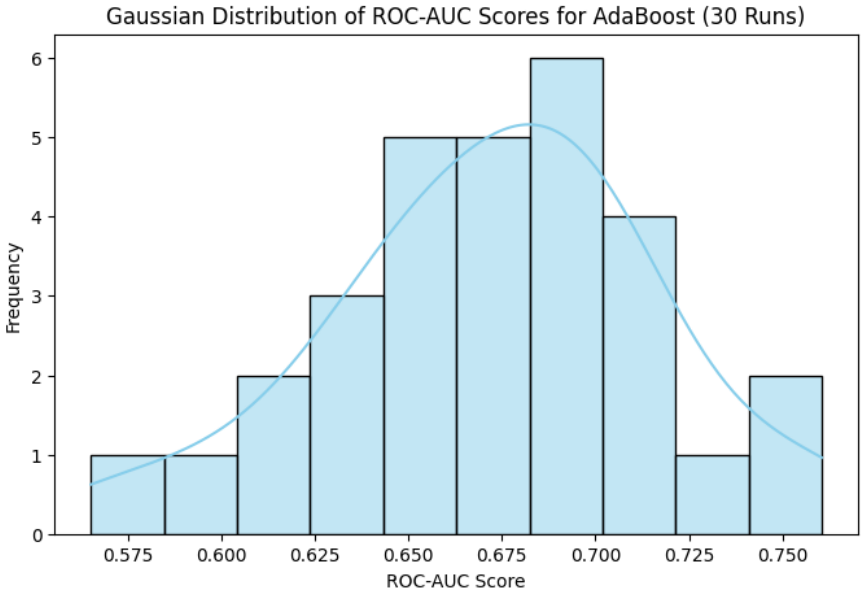


Figure 6

AdaBoost's scores show a high level of consistency, similar to Logistic Regression, with a narrow spread around the mean. This suggests that AdaBoost reliably improved performance by giving more weight to difficult cases during each iteration.

5.8 K- Nearest Neighbors

K-Nearest Neighbors (KNN) relies on the proximity of data points for classification, which made it less effective on this Gaussian-distributed dataset, as reflected in its lower ROC-AUC of 0.598. The nature of the Gaussian distribution, where data points may be closely clustered around the mean, likely caused overlap between classes, reducing KNN's ability to make clear distinctions[8]. Although its precision (0.661) was in line with other models, the lower recall (0.716) and F1 score (0.685) indicate that KNN struggled to identify mental health cases effectively (fig. 7). The model's performance suggests that while KNN can capture local patterns in well-separated data, it does not perform as well when classes are closely distributed, as seen in this study

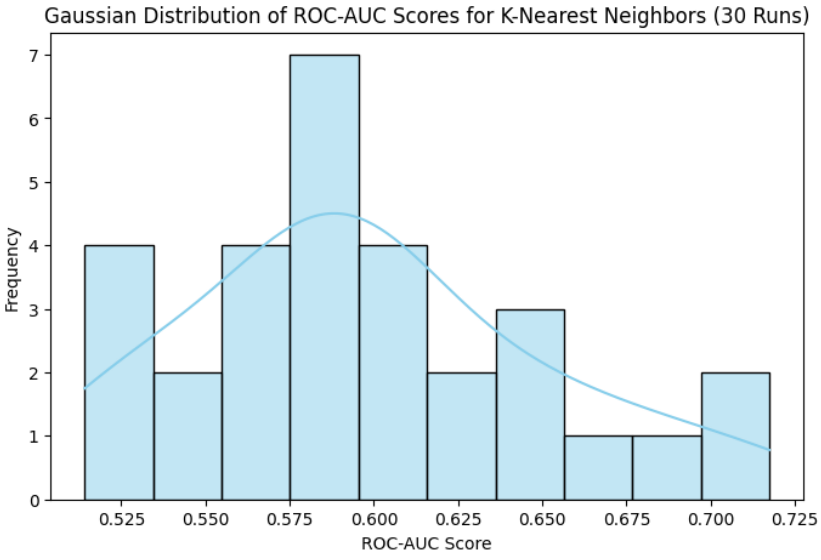


Figure 7

The KNN model's scores are more dispersed than those of the other models, indicating less stability in performance. KNN struggled with this dataset, leading to varying results depending on the specific data split in each run.

5.9 AUC-ROC Scores:

AUC-ROC (Area Under the Curve - Receiver Operating Characteristic) is a widely used performance metric for evaluating the effectiveness of binary classification models. It provides a single value that represents the model's ability to distinguish between the two classes, typically denoted as "positive" and "negative."

- **ROC Curve**: The ROC curve is a graphical plot that illustrates the diagnostic ability of a binary classifier by plotting the **True Positive Rate (Recall)** on the y-axis against the **False Positive Rate (1-Specificity)** on the x-axis at various threshold levels. A perfect classifier would have an ROC curve passing through the top-left corner, indicating high recall and low false positives.
- **AUC (Area Under the Curve)**: AUC represents the area under the ROC curve and summarizes the overall performance of the classifier. An AUC score ranges from 0 to 1, where:
  - **AUC = 1**: The classifier makes perfect predictions, distinguishing between classes without error.
  - **AUC = 0.5**: The classifier is performing no better than random guessing.
  - **AUC < 0.5**: The classifier is performing worse than random guessing, which may indicate a model that systematically predicts the wrong class.

The **AUC-ROC score** is especially useful in imbalanced datasets where accuracy alone might be misleading. By analyzing how well a model balances true positives and false positives across different threshold settings, AUC-ROC provides a more comprehensive picture of classification performance.

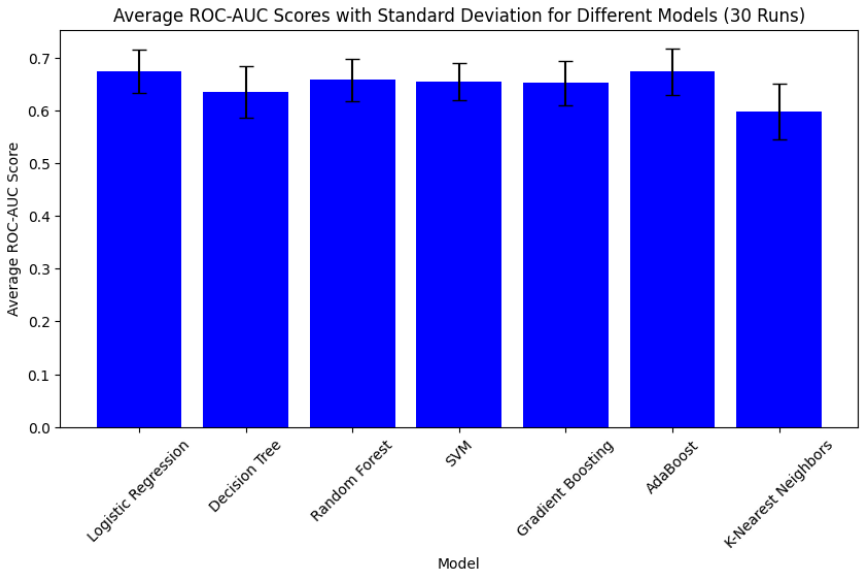


Figure 8

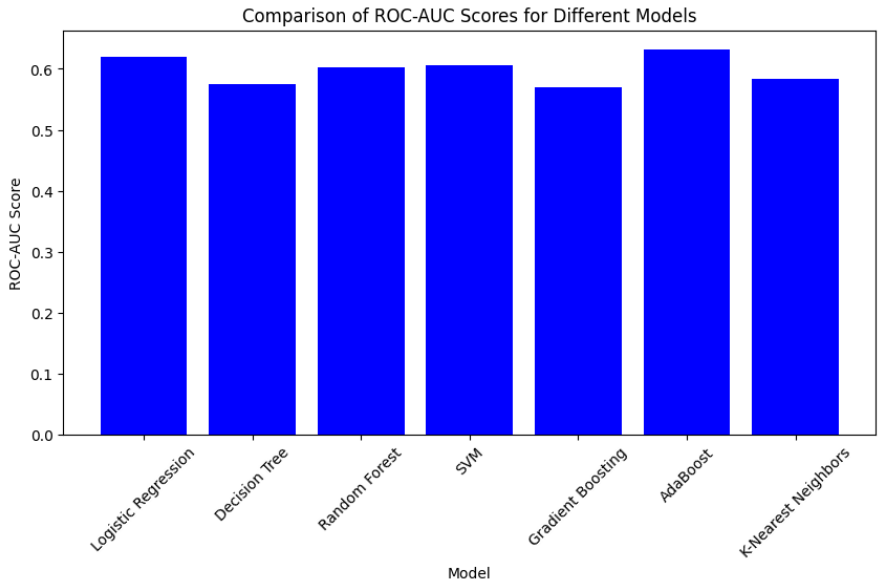


Figure 9

5.10 Comparison of Original vs Hypothetical scenario

For the comparison of ROC_AUC scores between the original and hypothetical scenarios, where certain significant features such as ‘Medu’, ‘Fedu’, and ‘studytime’ were removed, the analysis of how this change has affected each model’s performance is shown in Figure 10.

In the dataset, key features include "Medu" (Mother's Education), "Fedu" (Father's Education), and "studytime," which impact the prediction of mental health risks. Medu and Fedu are measured on a scale from 0 (none) to 4 (higher education), while studytime is categorized from 1 (less than 2 hours) to 4 (more than 10 hours). Although these features weren't directly used to generate the target variable (mental_health_issues), their removal in a hypothetical scenario showed a slight decline in ROC-AUC scores, indicating their positive contribution to the model's predictive performance. Their influence on students' academic and personal lives suggests they play a significant role in shaping mental health outcomes.

The Gradient Boosting model saw the largest decline in performance, indicating that it was the most dependent on the presence of all features, particularly the removed ones, to effectively make predictions.

The K-Nearest Neighbors (KNN) model actually improved when some features were removed, making it the least dependent on having all the features. This might suggest that KNN works better when the dataset has fewer, more relevant features.

Ensemble models like Gradient Boosting and AdaBoost were significantly impacted, while KNN benefitted from feature reduction. Gradient Boosting was the most reliant on the full feature set, whereas KNN was the least reliant, performing better when unnecessary or potentially noisy features were removed.

```
Comparison of ROC-AUC Scores (Original vs Hypothetical Scenario):
                     Original  Hypothetical
Logistic Regression  0.673580      0.665415
Decision Tree        0.632657      0.623450
Random Forest        0.654930      0.642235
SVM                  0.654873      0.646947
Gradient Boosting    0.656909      0.642308
AdaBoost             0.672900      0.662325
K-Nearest Neighbors  0.597538      0.605734
```

Figure 10

CHAPTER 6 EVALUATION

6.1 Single split vs 30 different splits:

- **Single Split**: This refers to training and testing the models on a single partition of the dataset. The models may be overfitting or underfitting, depending on how representative that single split is of the dataset as a whole. This single evaluation may give misleading performance metrics if the split is not fully representative of the overall data.
- **30 Runs (30 types of 80:20 splits)**: In this approach, the models are trained and tested over 30 different random splits of the dataset. By averaging the results over 30 different partitions, we get a more stable and reliable estimate of each model's performance. This method reduces the impact of randomness and provides better insights into the true generalization capability of the model.

1. ROC-AUC

Model	Single Split	30 Splits
Logistic Regression	0.6197	0.6736
Decision Tree	0.5887	0.6355
Random Forest	0.5743	0.6618
Support Vector Machine	0.6066	0.6549
Gradient Boosting	0.5711	0.6533
AdaBoost	0.6416	0.6729
K-Nearest Neighbors	0.5834	0.5975

Table 1

2. Accuracy

Model	Single Split	30 Splits
Logistic Regression	0.6400	0.6410
Decision Tree	0.5800	0.6210

Random Forest	0.5500	0.6374
Support Vector Machine	0.6200	0.6277
Gradient Boosting	0.5600	0.6213
AdaBoost	0.6000	0.6400
K-Nearest Neighbors	0.5900	0.5992

Table 2

3. Precision

Model	Single Split	30 Splits
Logistic Regression	0.7200	0.6635
Decision Tree	0.6900	0.6712
Random Forest	0.6200	0.6730
Support Vector Machine	0.6700	0.6772
Gradient Boosting	0.6300	0.6527
AdaBoost	0.6800	0.6646
K-Nearest Neighbors	0.6600	0.6611

Table 3

4. Recall

Model	Single Split	30 Splits
Logistic Regression	0.6700	0.8444
Decision Tree	0.5700	0.7741
Random Forest	0.6600	0.8023
Support Vector Machine	0.7100	0.7646
Gradient Boosting	0.6700	0.8285
AdaBoost	0.6600	0.8161
K-Nearest Neighbors	0.6700	0.7156

Table 4

5. F1-Score

Model	Single Split	30 Splits
Logistic Regression	0.6900	0.7413
Decision Tree	0.6200	0.7071
Random Forest	0.6400	0.7303
Support Vector Machine	0.6900	0.7144
Gradient Boosting	0.6500	0.7265
AdaBoost	0.6700	0.7419
K-Nearest Neighbors	0.6700	0.6854

Table 5

6.2 Statistical comparison of t-test p-value for original vs hypothetical scenario

t-test p-value

In the process of evaluating the models on the **original dataset** versus the **hypothetical scenario** (where features such as "Medu," "Fedu," and "studytime" were removed), a **paired t-test** was conducted to assess whether the performance difference in terms of ROC-AUC was statistically significant. The **p-value** from the t-test indicates whether the differences between the original and hypothetical scenario are statistically significant. For this analysis, a significance level of **0.05** (5%) was used. A p-value below this threshold indicates that the difference in performance between the original and hypothetical models is statistically significant.

• Logistic Regression

- **T-test p-value**: 0.0038
- The p-value is less than 0.05, indicating a **significant difference** in performance between the original and hypothetical scenarios. The removal of features had a noticeable impact on the Logistic Regression model.

• Decision Tree

- **T-test p-value**: 0.0083
- Similarly, the Decision Tree model showed a significant difference between the original and hypothetical datasets, as the p-value is below 0.05. The performance of the model was impacted by the removal of features.

• Random Forest

- **T-test p-value**: 0.1031
- The p-value is greater than 0.05, indicating that there is **no significant difference** in the performance of Random Forest between the original and hypothetical datasets. This suggests that the model is more robust to the feature changes and performed consistently across both scenarios.

• Support Vector Machine (SVM)

- **T-test p-value**: 0.0455
- The SVM model's p-value is slightly below 0.05, showing a **marginally significant difference** between the two datasets. This suggests that the feature removal had a minor impact on the model's performance.

• Gradient Boosting

- **T-test p-value**: 0.0452

- Similar to SVM, Gradient Boosting shows a marginally significant difference in performance, with a p-value just below 0.05. This indicates that the removal of features slightly affected its ability to classify the dataset.

• AdaBoost

- **T-test p-value**: 0.0139
- AdaBoost also showed a **significant difference** between the original and hypothetical scenarios. The feature removal impacted its classification performance, as reflected in the p-value.

• K-Nearest Neighbors (KNN)

- **T-test p-value**: 0.2319
- The KNN model has a p-value well above 0.05, suggesting **no significant difference** between the original and hypothetical datasets. This indicates that KNN was largely unaffected by the removal of the selected features.

The findings from the t-tests reveal that models like Logistic Regression, Decision Tree, AdaBoost, and SVM are sensitive to the removal of features, showing a significant performance change, while Random Forest and KNN were more robust to these changes. This suggests that some models rely heavily on specific features for accurate classification, while others are more resilient. Understanding these dependencies can guide feature selection and model choice in future applications, helping to improve model stability and generalization.

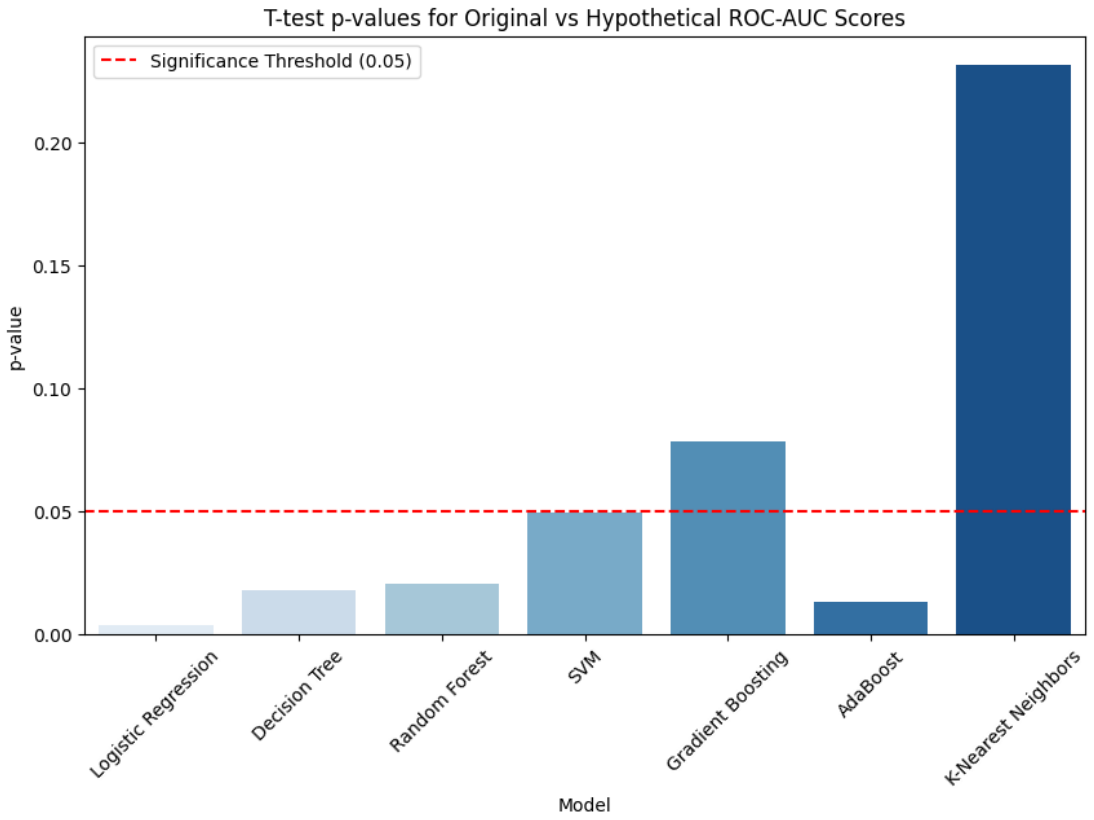


Figure 11

Model	T- test p-value	Significance (p<0.05)	Impact on Performance
Logistic Regression	0.0038	Yes	Significant impact; Feature removal affected performance
Decision Tree	0.0083	Yes	Significant impact; Feature removal affected performance
Random Forest	0.1031	No	No significant impact; performance largerly unaffected
Support Vector Machine	0.0455	Yes	Marginal impact; slight performance difference
Gradient Boosting	0.0452	Yes	Marginal impact; slight performance difference
AdaBoost	0.0139	Yes	Significant impact; Feature removal affected performance
K-Nearest Neighbors(KNN)	0.2319	No	No significant impact; performance largerly unaffected

Table 6

6.3 Prediction

In the evaluation of the machine learning models used for predicting mental health issues, a rigorous approach was followed to ensure the optimal model was selected based on its performance[3]. Preprocessing was done on the dataset, cleaning, normalising, and formatting it so that machine learning models could use it. To guarantee a fair and impartial assessment of each model's performance, the models were trained and tested on several runs with different data splits after the preprocessing stage.

The evaluation process was rigorous due to systematic preprocessing, including thorough cleaning and normalization of the dataset. Multiple data splits were used for training and testing, minimizing overfitting and ensuring reliable performance metrics. The ROC-AUC score served as a robust indicator of model effectiveness, allowing for objective model selection based on data-driven results. Visualizing predictions on the test dataset demonstrated the model's practical application in identifying at-risk students. This comprehensive approach highlights the importance of accuracy and dependability in mental health predictions.

Following the training and testing stages, a number of metrics were used to assess each model; the main performance indicator was the ROC-AUC (Receiver Operating Characteristic - Area Under Curve) score[7]. When evaluating a model's ability to differentiate across classes (in this case, students with and without mental health issues) the ROC-AUC score is a very trustworthy tool. The assessment made sure the selected model was reliable and efficient at spotting patterns of mental health hazards by utilising this metric.

The model with the highest ROC-AUC score was automatically determined to be the best-performing model after the results were analysed[10]. This was a critical step since it allowed the model that showed the best accuracy in identifying pupils with mental health issues from those without to be given priority by the system.

Following selection, the resulting model was used to visualise predictions on the test dataset. Out of all the students tested, 515 were expected to be at risk of developing mental health issues, whereas 134 students were predicted to have no mental health concerns, according to the significant prediction results. This distribution demonstrated how well the model identified a sizable portion of students who would benefit from mental health services.

The anticipated results show a distinct split in the student population and provide important information about how machine learning models might be used in educational environments to identify mental health issues early on. In order to understand how predictive analytics might be utilised to improve student well-being by identifying those who can benefit from timely interventions, this evaluation is a necessary first step[12]. It also emphasises how crucial it is to choose the appropriate model for these delicate activities because the precision and dependability of forecasts have a direct bearing on actual results.

Logistic Regression and AdaBoost emerged as the top-performing models, both achieving high ROC-AUC scores around 0.673, indicating their effectiveness in distinguishing between students at risk and not at risk for mental health issues. Their strong precision and recall across multiple runs demonstrate their reliability in identifying patterns, making them optimal choices for this dataset. In contrast, Random Forest and Gradient Boosting performed well but did not surpass these models in ROC-AUC scores. K-Nearest Neighbors showed the lowest performance, struggling with the dataset's complexity.

The ROC-AUC score in particular is the focus of this study of the data, which highlights the importance of accuracy and comprehensiveness in the evaluation of models for mental health prediction because inaccurate forecasts might have serious repercussions fig. 12.

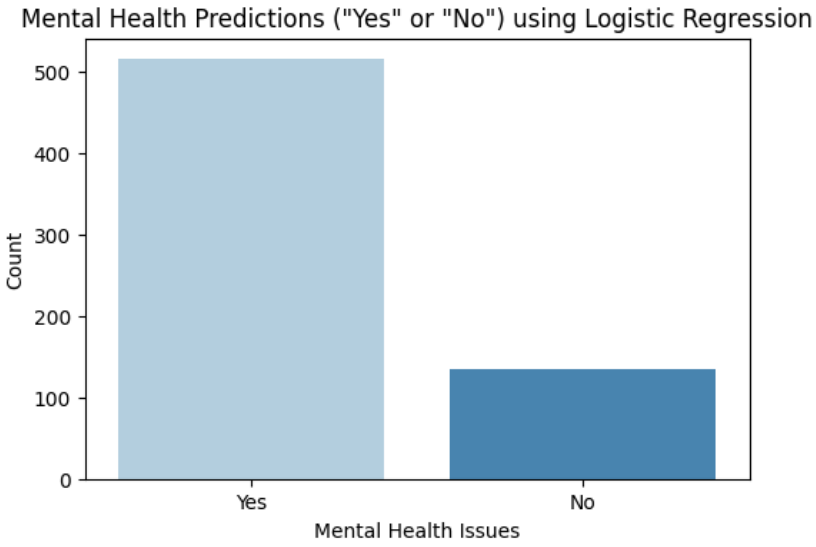


Figure 12

CHAPTER 7 CONCLUSION AND FUTURE WORK

7 7.1 Conclusion

- **Logistic Regression** and **AdaBoost** tend to perform the best, both achieving high ROC-AUC scores, precision, and recall across multiple runs. The ROC-AUC score for these models is around **0.673**, which indicates good overall

performance in distinguishing between students at risk and not at risk of mental health issues.

- **Random Forest** and **Gradient Boosting** also perform well, but slightly below Logistic Regression and AdaBoost. These models are generally robust and can handle different types of data well, but they might not be the optimal choice for this dataset as they don't outperform in terms of ROC-AUC.
- **K-Nearest Neighbors (KNN)** showed the lowest ROC-AUC score and struggled with more complex data, making it the least suitable for this dataset.

Robust performance measurements were employed to train, test, and assess a variety of models. The principal metric employed to guarantee dependable and precise predictions was the ROC-AUC score[3]. Using technology for early intervention and assistance in educational environments has advanced significantly with the model's ability to identify children who may be at risk for mental health issues.

However, the model's capacity to generalise to bigger and more varied populations is limited because the forecast was based on a dataset with only 649 occurrences. As a matter of professional concern, no other publicly accessible dataset presently possesses the same features that were employed in this effort[4]. The model's potential for wider applicability is hampered by this lack of data availability. Access to a more extensive and larger dataset is essential for improving the model's training efficiency and prediction accuracy[9]. Better predictions would result from the model being able to include a larger range of behaviours and traits that raise the risk of mental illness.

Furthermore, inferences from the available data were used to define the threshold values for the target variable in the mental health forecasts[14]. For instance, criteria were manually set based on perceived significance for alcohol usage, health status, and academic failures. Although the current model was built on these assumptions, they could not be the best for datasets or people in the future[7]. The threshold values should be reassessed and perhaps modified as more data become available in order to better represent the features of the new dataset. This would guarantee the model's applicability to a wider range of people in addition to enhancing forecast reliability.

7.2 Future Work

1. **Access to Larger and Diverse Datasets**: The immediate next step should be the collection or acquisition of a larger and more diverse dataset to improve the model's generalization capability. This can be achieved through collaborations with educational institutions or by utilizing advanced data collection techniques.

2. **Dynamic Threshold Adjustment**: Future versions of the model should incorporate dynamic methods for adjusting the threshold values used to set the target variable. This would enhance the model's adaptability to new datasets, leading to more accurate and reliable predictions.

3. **Improved Feature Engineering**: Additional features related to behavioral, psychological, and socio-economic factors should be explored to increase the model's predictive accuracy. By enriching the feature set, the model can better capture the complexities of mental health issues.

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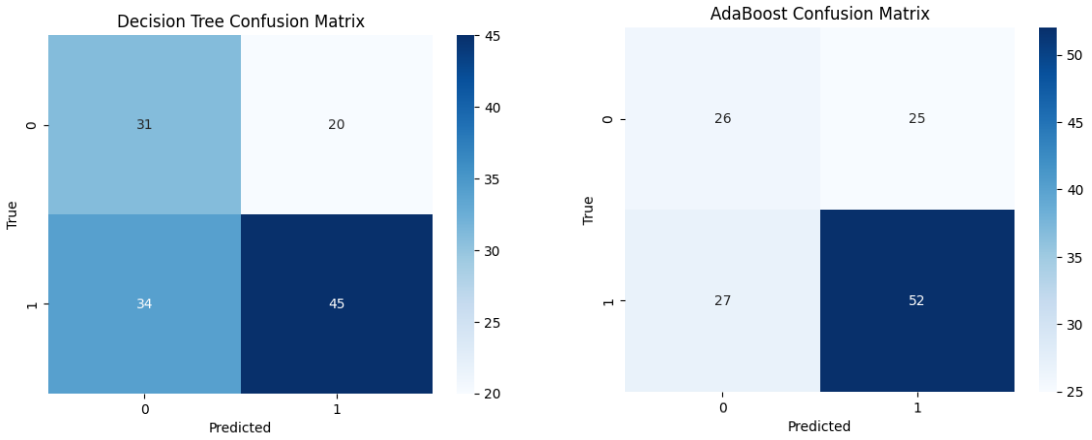
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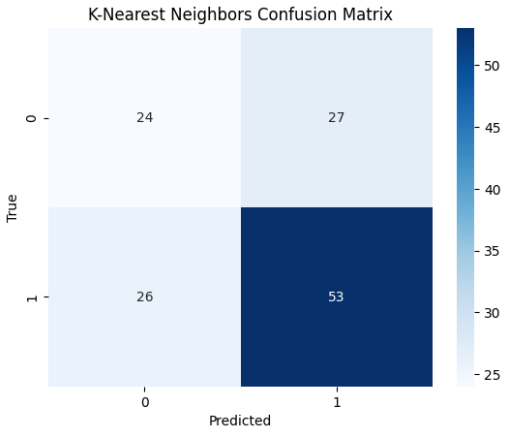
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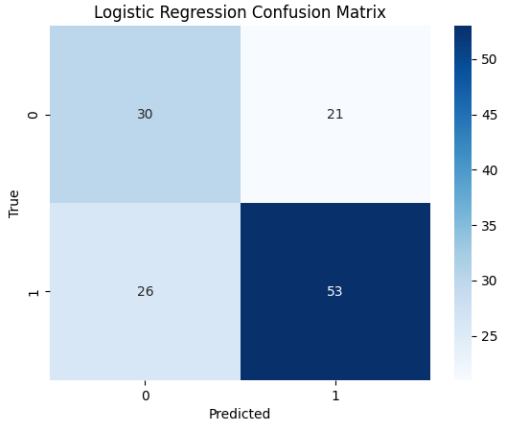
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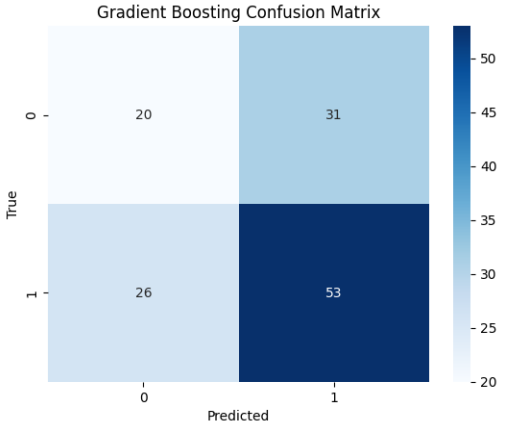
CHAPTER 10 APPENDICES

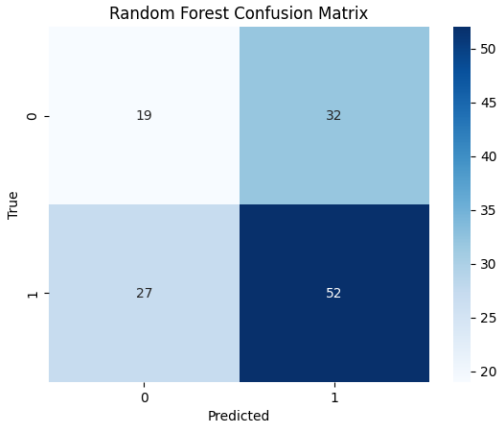
APPENDIX A. CONFUSION MATRIX

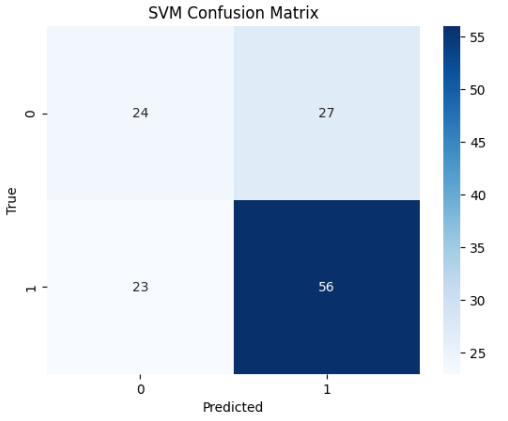












Latest version: <**http://TROSTing.org/info/2005/08/i050803b.htm**> provides the latest material on the TROST Pattern Description Approach.

Version: 0.30 <**http://TROSTing.org/info/2005/08/i050803e.htm**> is the version of the approach that this appendix is based on.

10.1 Gaussian Distribution

Gaussian distribution graphs represent the normal distribution of a dataset. In these diagrams, the curve shows the probability density function, with the center (mean) being the highest point of the curve. The x-axis represents values, while the y-axis represents the probability density. The bell shape illustrates the data's spread, with lines marking one, two, and three standard deviations away from the mean. This allows an understanding of how much of the data falls within those ranges.